

CS5489

MACHINE LEARNING: ALGORITHMS & APPLICATIONS

LECTURE 1 - INTRO

2026/27 Semester A

Course General Info

□ Teaching Team

▣ Prof. Jiawei Ma and Prof. Zhisong Zhang

■ jiaweima@cityu.edu.hk (AC1-Y6426), zhisong.zhang@cityu.edu.hk (AC1-Y6613)

▣ TAs: Announced in Canvas

▣ Office hours: Announced *in Canvas*

□ Canvas-based course site

- ▣ It is your own responsibility to check Canvas Announcement and University e-mail account regularly for announcements and updates.
- ▣ Raise your questions under Discussion.

Class Times (Tuesdays)

Daytime

(Prof. Zhisong ZHANG)

□ Lecture

▣ 12:00-13:50 (LT-2)

□ Tutorials

▣ 14:00-14:50 (B7701)

▣ 15:00-15:50 (B7701)

Evening

(Prof. Jiawei MA)

□ Lecture

▣ 19:00-20:50 (LT-2)

□ Tutorials

▣ 18:00-18:50 (B7701)

▣ 21:00-21:50 (B7701)

Note: No tutorial in the first week

*: Tutorial Participation Survey Needed for Each week

Teaching Activities

- **Lectures** (2 hours per week)
 - ▣ present machine learning algorithms: intuition and idea, and algorithm. Illustrate algorithms on both toy and real-world examples.
- **Tutorials** (1 hour per week)
 - ▣ Use machine learning algorithms on small examples to gain better understanding. Implement algorithms.
- **Assignments** (2)
 - ▣ Apply machine learning algorithms to larger datasets, compare and interpret the results of different algorithms.
- **Course Project**
 - ▣ Apply machine learning to solve a real-world problem.
 - Computer Vision & Natural Language Processing (self-directed project with approval).
 - ▣ 2-3 students per group (cross-session is allowed, due in Week 3).

Assessment

□ **Coursework (70%)**

- Tutorial exercises (10%) – due 2 weeks after lecture (Tue).
- Assignments (20%) – due Weeks 5 and 9 (Fri).
- Midterm (10%) – Week 7
- Course Project (30%) – due Week 14
 - Grouping is due Week 3
 - Project proposal, report, and presentation.

□ **Final Exam (30%)**

- **Note:** Must get at least 30% on final exam and 30% on course project to pass the course.

Programming

□ Python

- ▣ high-level scripting language
- ▣ Jupyter, aka iPython Notebooks (ipynb)
 - interactive computational environment in web browser.
 - can combine text (Markdown) with Python code and output.
- ▣ Libraries
 - numpy – arrays, linear algebra
 - scikit, scikit-learn – scientific computing, machine learning
 - matplotlib, pylab – plotting
 - Pytorch / keras – deep learning
- ▣ Introduction later today

Assignments

□ **Assignment (Individual)**

- ▣ How deep learning is used to solve a problem.
- ▣ How the deep neural network is updated.

□ **Projects (Teamwork)**

- Topic candidates: to be announced in canvas.
- Strongly encourage and welcome self-directed project
 - A project proposal is required to enable our team to provide the guidance and resources needed to ensure your project's success.

□ **Code/Report submission**

- Jupyter notebooks (ipynb)
- Python scripts (for project, if necessary)

Sitting in ...

- *Can I just attend the lectures without registering?*
- Reasons to take the course:
 1. The assignments are also teaching activities that give you hands-on experience (both in theory and programming).
 - If you are going to do these, you might as well do it for credit.
 2. Free to do whatever you want for the course project – apply ML to your research problem / ideas.
 - A good course project could lead to a publication.
 3. Machine Learning is an important topic in CS and increasingly important in other fields.
 - It looks good on your transcript/CV.
- Space in lecture hall is limited...
 - Seats are reserved for students who have registered for the course (or are planning to register).

Course Abstract

- The goal of this course is to introduce students to the field of machine learning, its algorithms and applications.
 - ▣ Machine learning algorithms allow computers to automatically learn to recognize complex patterns from empirical data, such as text and web documents, images, videos, sound, sensor-data, and databases.
 - ▣ This course is intended to give a broad overview of machine learning from the practical standpoint, with a focus on implementing and applying machine learning algorithms to real-world problems.
 - ▣ At the end of the course, students will have both working knowledge of and practical experience implementing and applying machine learning algorithms on different domains.

CILOs

1. Identify and explain common machine learning algorithms.
2. implement machine learning algorithms
3. Apply machine learning algorithms to solve real-world problems.
4. Evaluate the effectiveness of different machine learning algorithms and discuss their advantages and disadvantages.

What is Machine Learning?

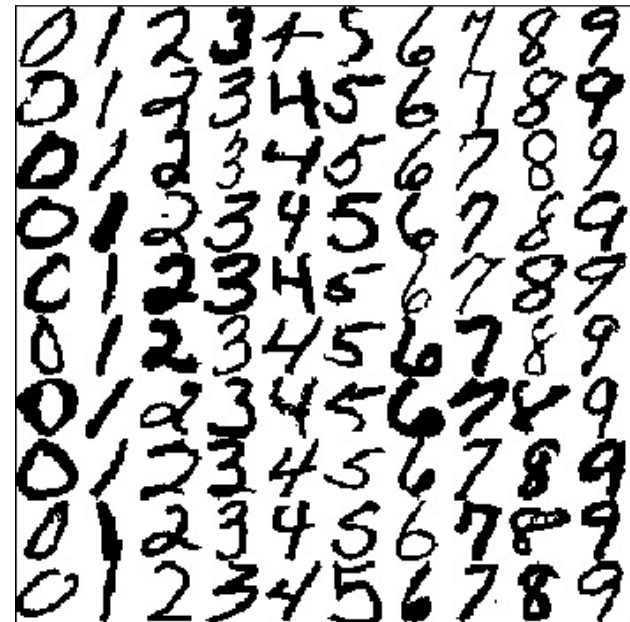
- Arthur Samuel, 1959

- ▣ *Machine Learning*: field of study that gives computers the ability to learn without being explicitly programmed.

- e.g. computer learns to play checkers by playing against itself.

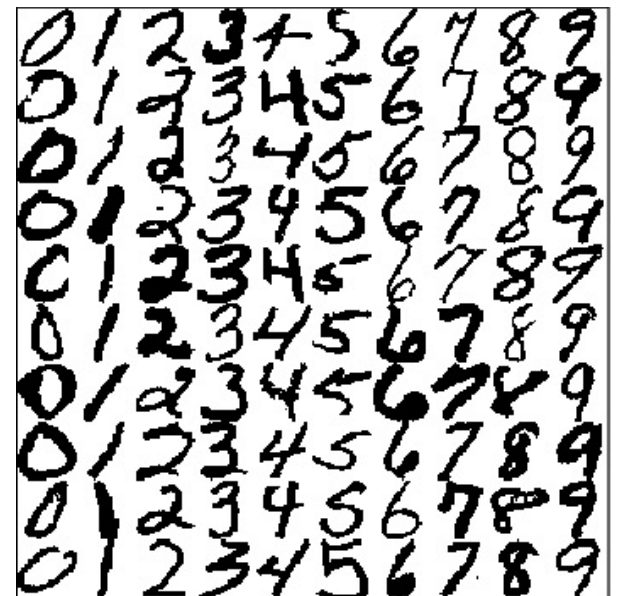
- There are many applications that are difficult to program by hand.

- ▣ Example: Recognizing handwritten digits in an image.



What is Machine Learning?

- Example: Recognizing handwritten digits in an image
 - ▣ 28x28 image $\rightarrow$ 784-dim vector
 - ▣ a lot of variations & permutations
 - ▣ difficult to identify rules & code by hand
- ML solution:
 - ▣ gather some example data.
 - ▣ train computer to discover differences automatically



What is Machine Learning?

- Tom Mitchell, 1997
 - ▣ *Well-posed Learning Problem*: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."
 - e.g., the computer gets better at recognizing digits as it sees more examples, as measured by the error rate.
- A closer look...

Well-posed Learning Problem

- "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."

"class of tasks T "

learning is
task-specific
(recognition,
clustering, etc.)

"performance measure P "

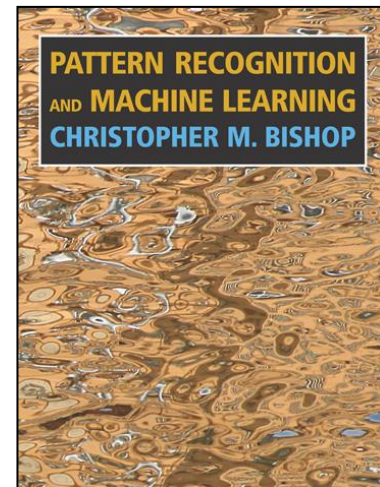
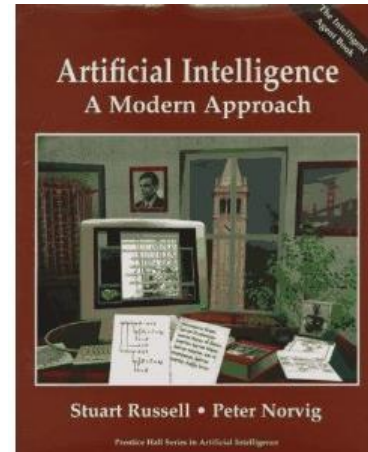
optimize a loss function
(e.g., error rate), but also
prevent overfitting
(regularization).
"generalization"

"experience E "

data-driven!
More data is
better!

Machine Learning vs. Artificial Intelligence

- Machine learning grew out of early work in AI
 - ▣ and other fields: statistics, physics, neuroscience, ...
 - ▣ fueled by more powerful computers and more data.
- "Traditional" Artificial Intelligence (Russell-Norvig)
 - ▣ Turing test (is it a computer or a human?)
 - ▣ solving by searching (A^* , α - β pruning, game playing)
 - ▣ knowledge-based (representation, reasoning, logic)
 - ▣ planning, scheduling, natural language processing
- Machine Learning (Bishop)
 - ▣ probability, statistics, Bayesian formulation
 - ▣ statistical learning theory
 - ▣ regression, classification, clustering



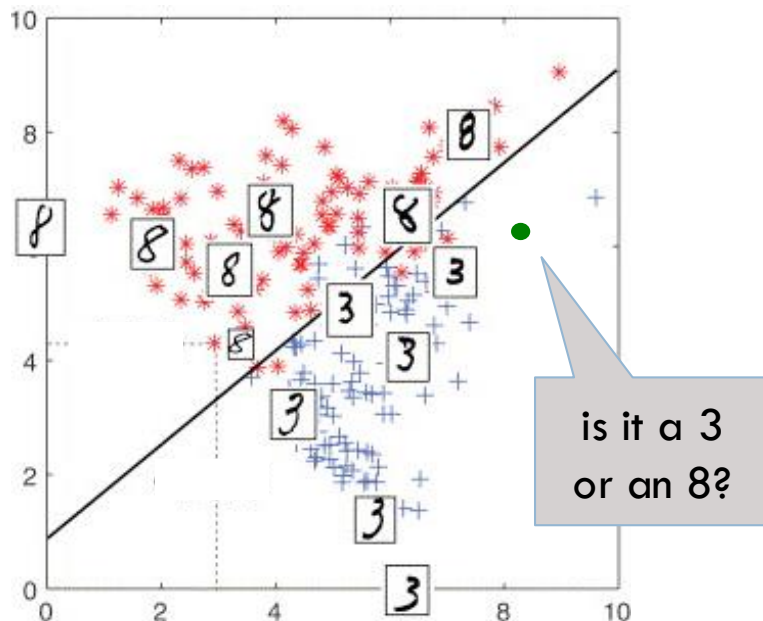
Topics in Machine Learning

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Learning Theory
- Deep Learning

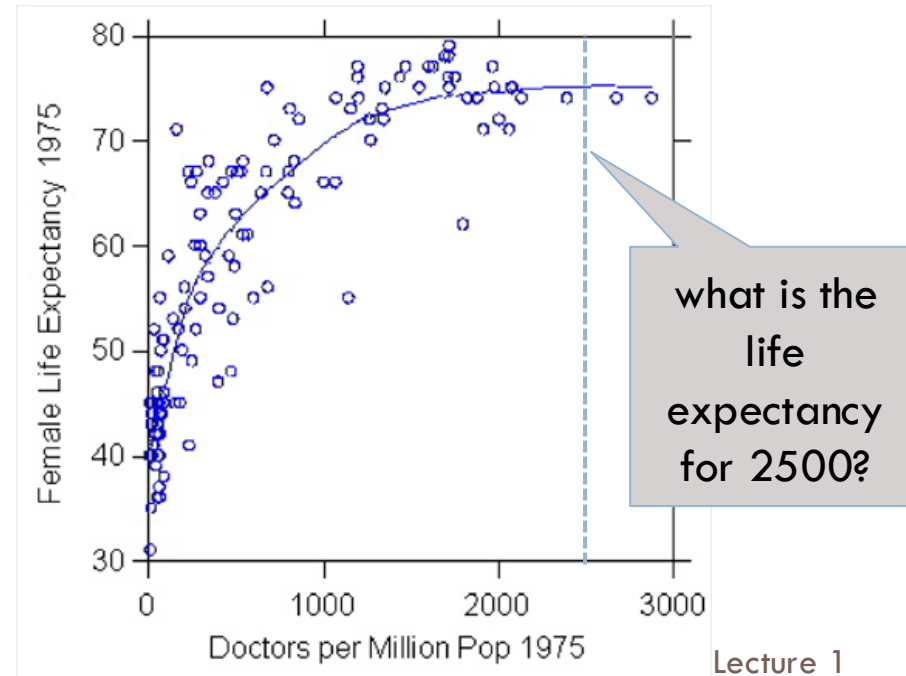
Supervised Learning

- Training data has inputs and outputs
 - ▣ e.g., digit recognition (input=image, output=digit)
 - ▣ learn a function mapping inputs to outputs

Classification

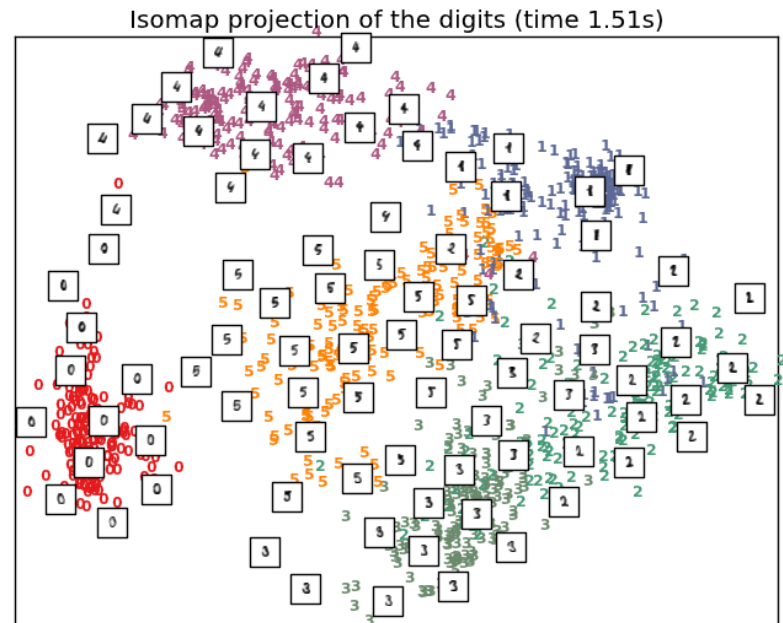
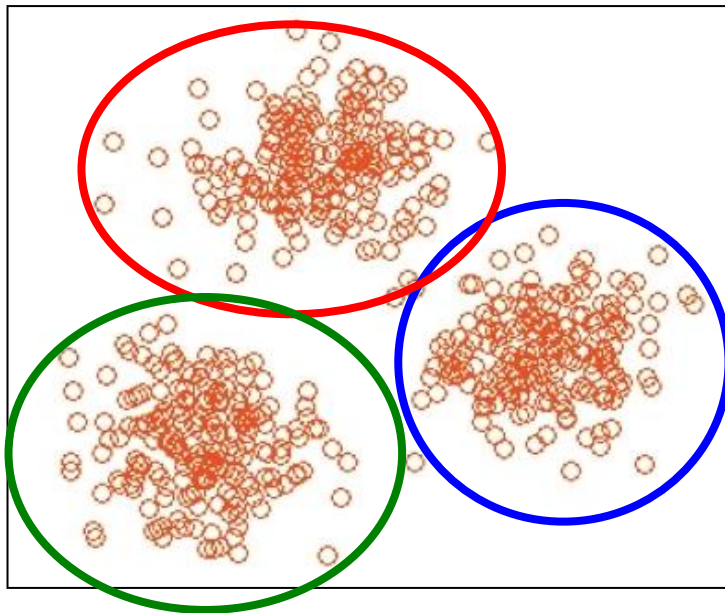


Regression



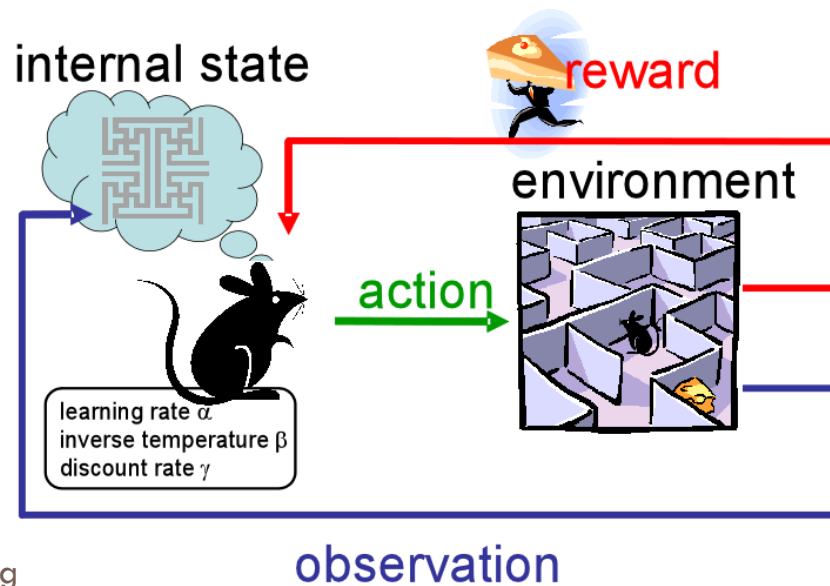
Unsupervised Learning

- Training data only has inputs (no outputs)
 - ▣ e.g., collection of web documents
 - ▣ clustering - discover groups of similar examples.
 - ▣ visualization - project high-dim data to 2 or 3-dimensions.



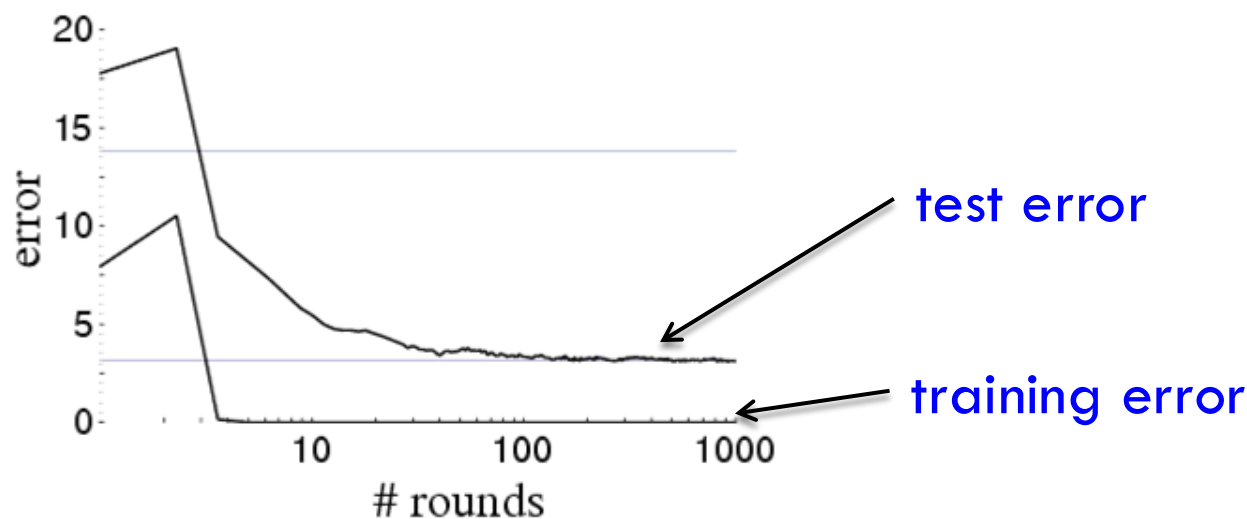
Reinforcement Learning

- Make a sequence of actions, given current states
 - ▣ e.g. a robot interacting with its environment
 - ▣ Maximize the reward
 - at some point, receive a reward or a punishment.
 - actions may also affect future reward.



Learning Theory

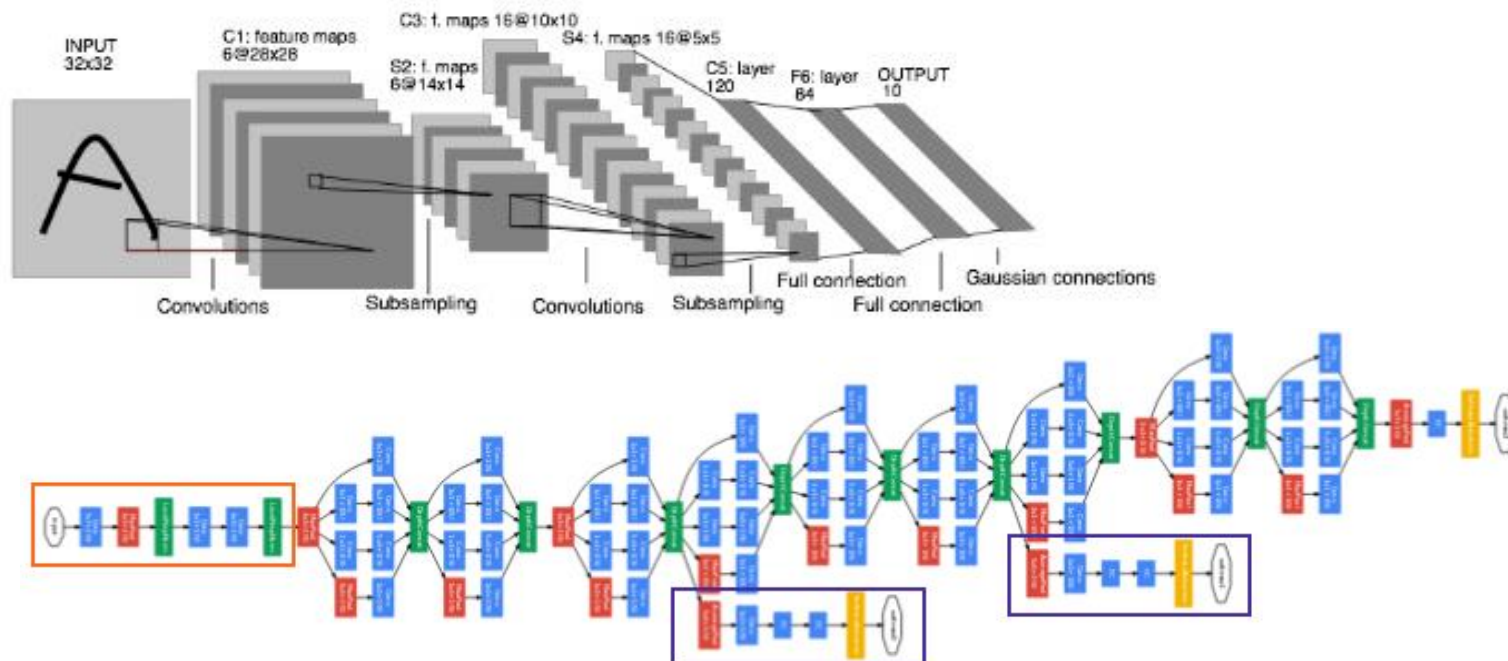
- Why does machine learning work?
 - ▣ performance guarantees – bounds on the expected test error.
 - ▣ What types of functions can be represented by an algorithm, and how much data do we need?



Deep Learning

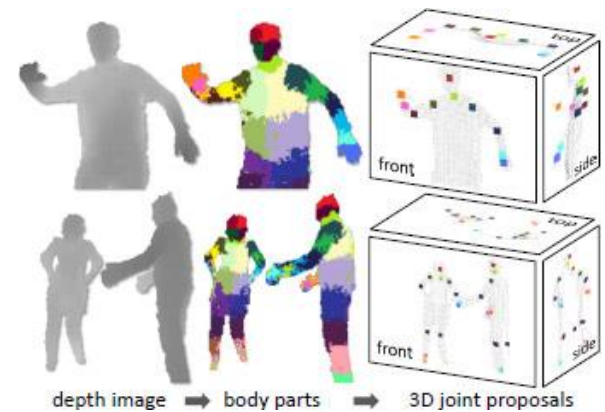
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- Deep learning is supervised/unsupervised learning using multi-layer neural networks.
 - ▣ Improved training algorithms to prevent overfitting
 - ▣ Faster parallelization (GPUs)
 - ▣ More data (millions of examples)

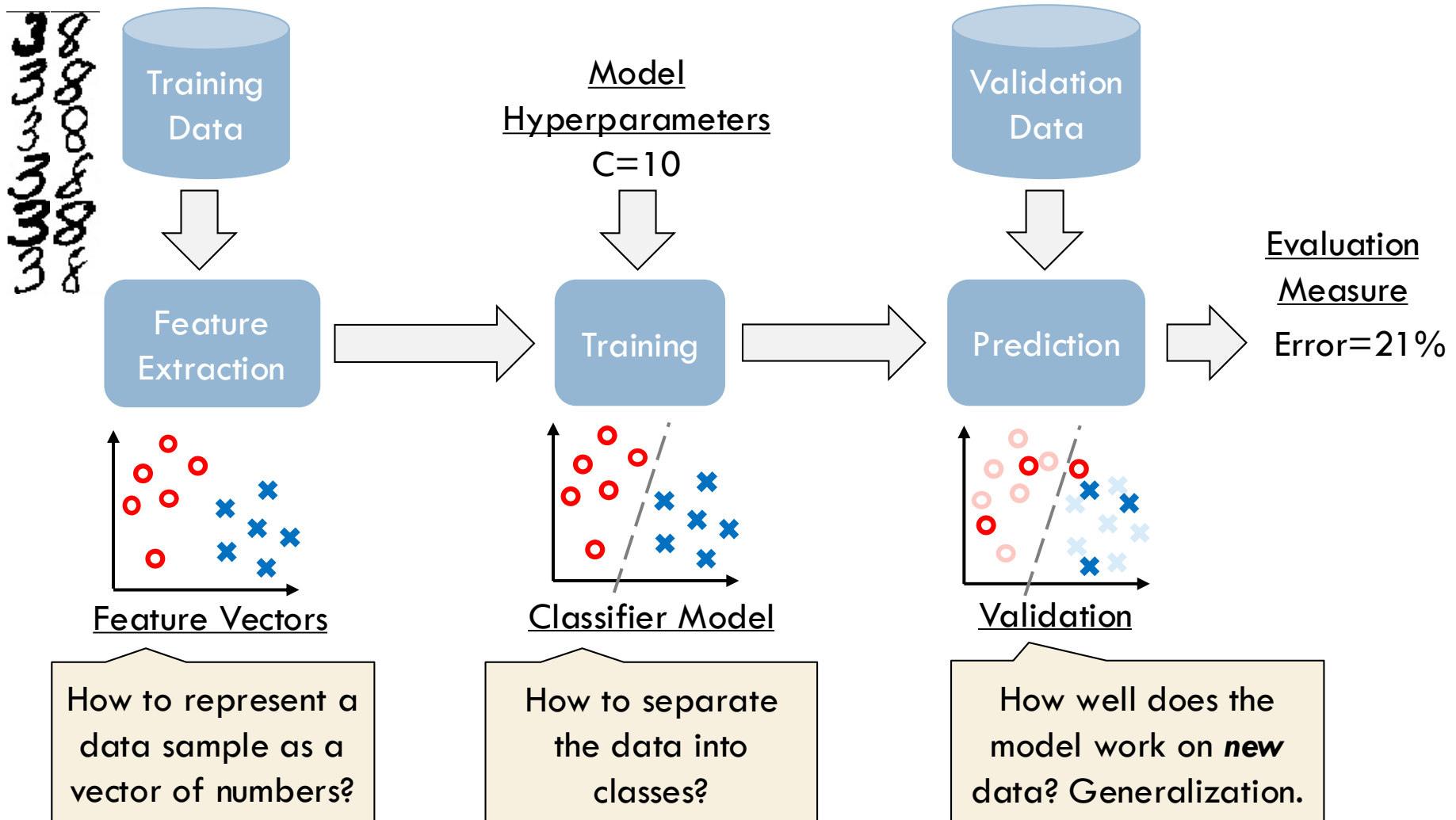


ML in the Real World

- Google
 - ▣ spam email classifier, speech recognition, AlphaGo, machine translation, image search, quick links,
- Face detection & recognition
 - ▣ digital cameras, Google street view, Facebook
- Business
 - ▣ credit card fraud detection
 - ▣ stock trading (portfolio optimization)
- Recommendation systems
 - ▣ Netflix, Amazon
- Human pose recognition (Kinect)
- Controllers (reinforcement learning)
- Self-driving Cars
- Generative AI (ChatGPT, Dall-E)

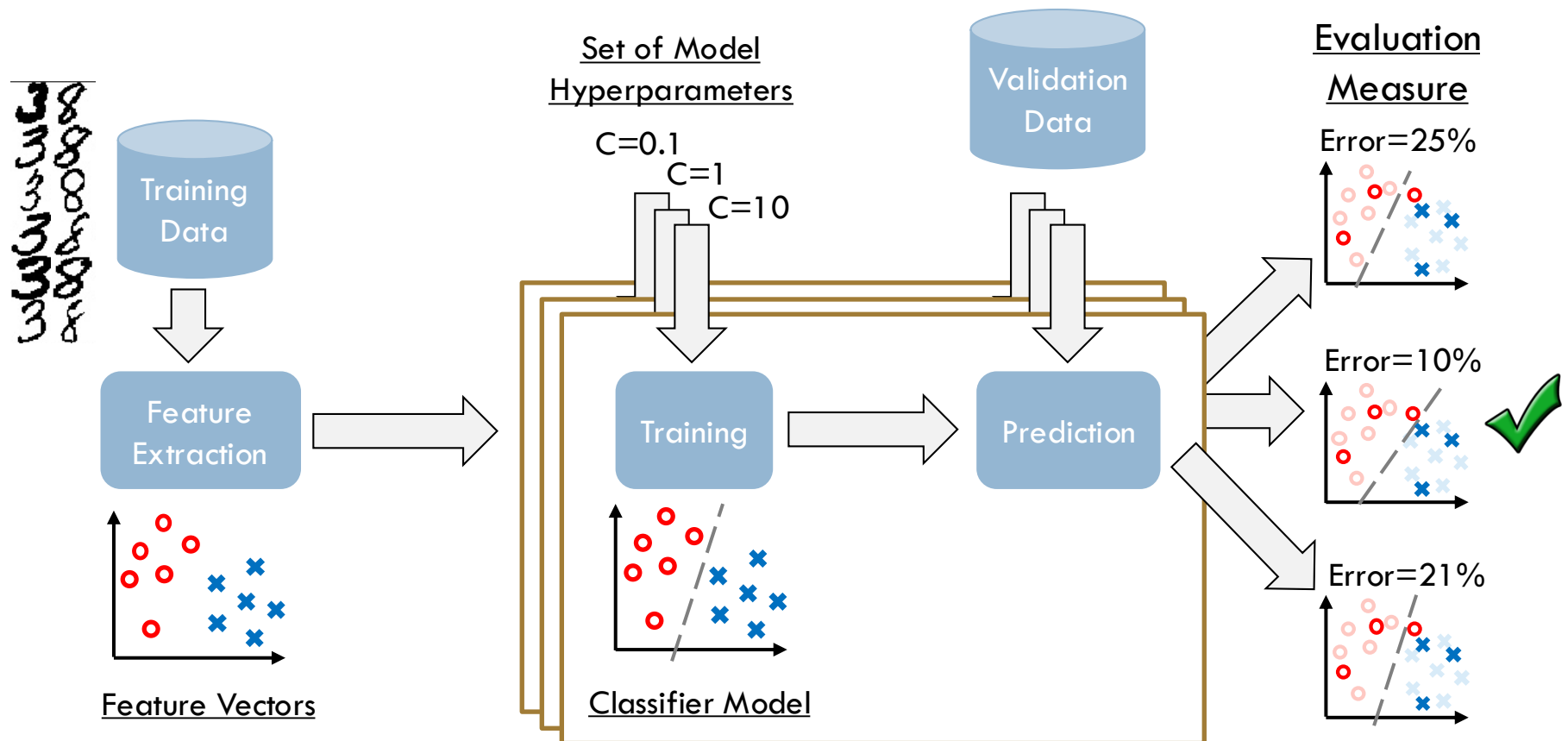


ML Training Pipeline



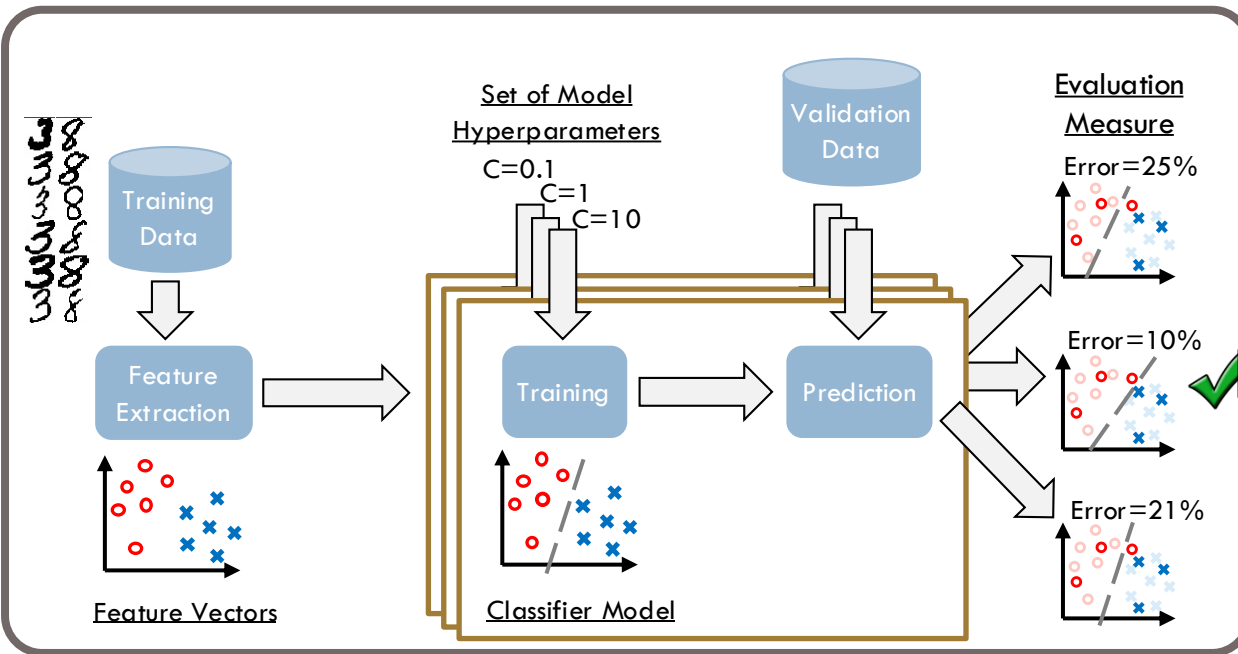
Model Selection Pipeline

- Select hyperparameters that yield low generalization error.



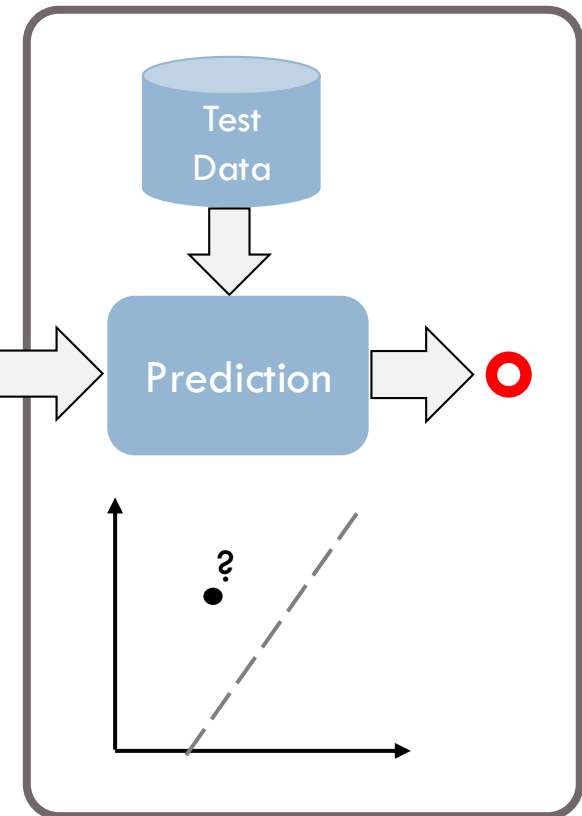
Training & Inference Stages

Training Stage



Typically slow, because need to try different hyperparameters and iterative over a training algorithm.

Testing (Inference) Stage



Typically fast, because the model is already estimated.

Building Blocks for ML

- What are the tools needed for Machine Learning?
 - ▣ *linear algebra*
 - matrices, inverse, eigenvector, SVD, ...
 - ▣ *probability & statistics*
 - random variables, expectation, Bayes' theorem, ...
 - ▣ *optimization algorithms*
- Don't worry, we will review these as necessary.
 - ▣ we need these tools to understand *how* ML works, and implement ML algorithms.
 - ▣ many algorithms are already implemented in the ML libraries.

THIS IS YOUR MACHINE LEARNING SYSTEM?

YUP! YOU POUR THE DATA INTO THIS BIG PILE OF LINEAR ALGEBRA, THEN COLLECT THE ANSWERS ON THE OTHER SIDE.

WHAT IF THE ANSWERS ARE WRONG?

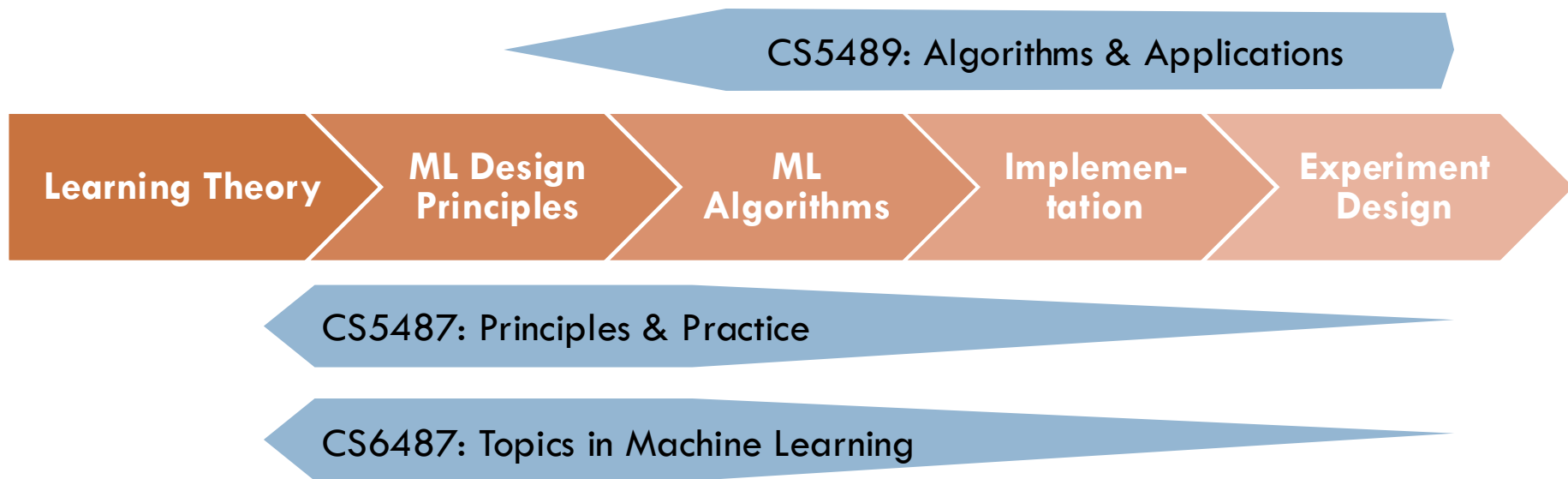
JUST STIR THE PILE UNTIL THEY START LOOKING RIGHT.



<https://xkcd.com/1838/>

Relationship with other ML courses

- CS5489 mainly focuses on the intuition of how machine learning algorithms work, implementation of algorithms, applying machine learning and analyzing the results.



- CS5487/CS6487 mainly focus on ML design principles and derivation of algorithms.
- CS5491 (AI) covers knowledge representation, uncertainty reasoning, reinforcement learning, search etc.

CS5489 Schedule (2026A)

Week	Lecture Topic (2025)	Reference				Tutorial	Assessment
		MR (pg)	H (ch)	B (ch)	GBC (ch)		
1 (Sep 1)	Lec 1: Introduction / Python <u>Supervised Learning</u>						Pr. out
2 (Sep 8)	Lec 2: Probabilistic Models & Bayes Classifiers	70-71	4	4.2		Tut 1	A1 out
3 (Sep 15)	Lec 3: Discriminative Classifiers & Nonlinear Classifiers	58-70, 85-106	5, 6, 7	4.3, 7.1, 1.4.2, 1.4.3		Tut 2	Grp. due,
4 (Sep 22)	Lec 4: Regression <u>Deep learning</u>	47-58	8, 9	3.1, 7.1, 6.4		Tut 3	
5 (Sep 29)	Lec 5: Neural networks	106-201		5.1-5.5	6	Tut 4	A2 out (Tus), A1 due (Fri)
6 (Oct 6)	Lec 6: CNN			5.5	7, 9	Tut 5	
7 (Oct 13)	Midterm (Lec 2-5)						
8 (Oct 20)	Lec 7: Deep Learning (No jokes!) and Its Problems <u>Unsupervised Learning</u>				8	Tut 6	(Optional) Proposal Due
9 (Oct 27)	Lec 8: Dimension Reduction	142-170	13, 14	12		Tut 7	A2 due (Fri)
10 (Nov 3)	Lec 9: Clustering <u>Advanced Deep Learning</u>	170-183	10.1-10.4	9.1-9.3		Tut 8	
11 (Nov 10)	Lec 10: Deep generative models				14, 20.9, 20.19	Tut 9	
12 (Nov 17)	Lec 11: LLM					Tut 10	
13 (Nov 24)	Lec 12: VLM						Pr. Pres. & Pr. Due (Fri)

References

□ Textbooks

- **MG:** Muller & Guido, “Introduction to Machine Learning with Python”, O’Reilly, 2017.
- **H:** Harrington, “*Machine Learning in Action*”, Manning Publications Co., 2012.
- **B:** C.M. Bishop, “*Pattern Recognition and Machine Learning*”, Springer, 2006.
- **GBC:** Goodfellow, Bengio, Courville, “*Deep Learning*”, MIT Press 2016.
 - <http://www.deeplearningbook.org>

□ Papers:

- Batch Norm: <https://arxiv.org/abs/1502.03167>
- SGD: <https://arxiv.org/abs/1802.06175>
- ResNet Ensembles: <https://arxiv.org/abs/1605.06431>

Other References

□ Online Reference Books

- ▣ A. Rajaraman, and J. Ullman, “*Mining of Massive Datasets*”, Cambridge University Press, 2011. (<http://infolab.stanford.edu/~ullman/mmds.html>)
- ▣ H. Daume III, “*A course in Machine Learning*”, (<http://ciml.info/>)

□ Other Books

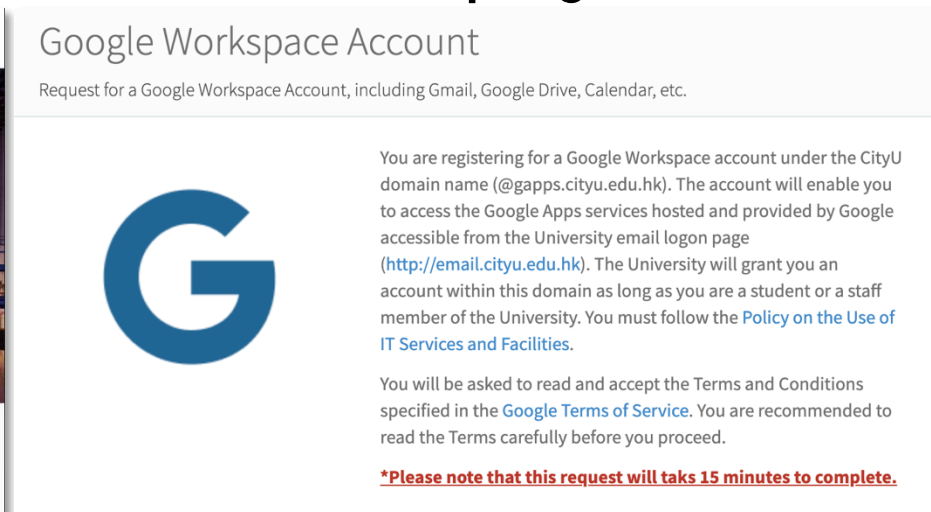
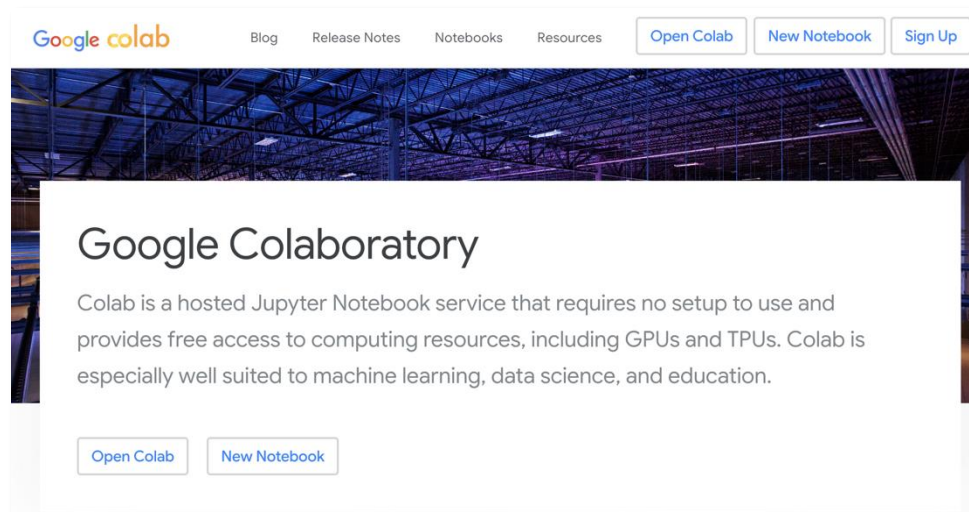
- ▣ R.O. Duda, P.E. Hart, & D.G. Stork, “*Pattern Classification (2nd Ed.)*”, Wiley-Interscience, 2001.
- ▣ T. Hastie, R. Tibshirani, and J. Friedman, “*The Elements of Statistical Learning: Data Mining, Inference, and Prediction (2nd Ed.)*”, Springer-Verlag, 2009.

Computing Resources (CPU)

□ Google Colab

- ▣ Provides Jupyter notebooks via CityU Google Account.
- ▣ Select between CPU and GPU (not-guaranteed) resource
- ▣ *dedicated for GPU-free course practices.*

□ More details will be shared via Canvas page.



Computing Resources (GPU)

□ **Third-Party Cloud**

- Account information will be given after grouping.
 - GPU-Guaranteed
 - Select between CPU and GPU resource
- More details will be shared via Canvas page.

Academic Honesty

- CityU has *Rules of Academic Honesty* and has required all students to complete an online tutorial on subject and declare your understanding
- Plagiarism...
 - ▣ It is serious fraud to plagiarize others' work.
 - ▣ Punishment ranges from warning to course failure.
- How to prevent plagiarism...
 - ▣ Finish the assignments by yourself! You have to write the program/solution yourself.
 - okay to talk about *how* to do the problem with your classmates.
 - **Protect your code**; don't give it away as a "reference" copy.
 - ▣ In plagiarism cases, we treat the giver and the copier as both guilty.
 - ▣ You hurt your own grades by not reporting cheating.
- As instructors...
 - ▣ We have responsibility to report academic dishonesty cases so as not to compromise the quality of education.
 - ▣ We take suspected plagiarism cases very seriously.

Machine Learning



what society thinks I
do

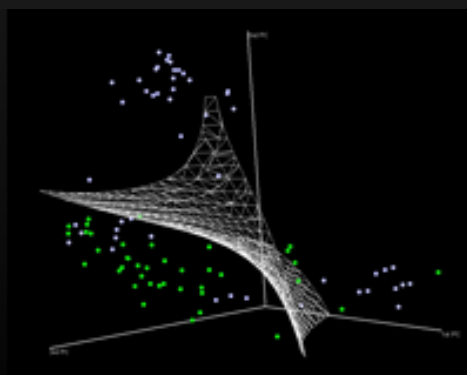


what my friends think
I do



what my parents think
I do

$$\begin{aligned}
 L_p &= \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^n \alpha_i y_i (\mathbf{x}_i \cdot \mathbf{w} + b) + \sum_{i=1}^n \alpha_i \\
 \alpha_i &\geq 0, \forall i \\
 \mathbf{w} &= \sum_{i=1}^n \alpha_i y_i \mathbf{x}_i, \sum_{i=1}^n \alpha_i y_i = 0 \\
 \nabla \hat{g}(\theta_t) &= \frac{1}{n} \sum_{i=1}^n \nabla \ell(x_i, y_i; \theta_t) + \nabla r(\theta_t) \\
 \theta_{t+1} &= \theta_t - \eta_t \nabla \ell(x_{i(t)}, y_{i(t)}; \theta_t) - \eta_t \cdot \nabla r(\theta_t) \\
 \mathbb{E}_{i(t)}[\ell(x_{i(t)}, y_{i(t)}; \theta_t)] &= \frac{1}{n} \sum_i \ell(x_i, y_i; \theta_t)
 \end{aligned}$$



```
>>> from scipy import SVM
```

what other programmers
think I do

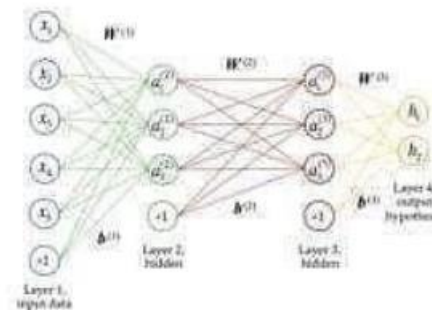
what I think I do

what I really do

A parrot



Machine learning algorithm



Learns random phrases



Doesn't understand anything about what it learns



Occasionally speaks nonsense



Is a cute birdie parrot

